INCH-POUND

MIL-S-13949/5D
6 August 1993
SUPERSEDING
MIL-S-13949/5C
28 March 1989

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL GH
(WOVEN E-GLASS REINFORCEMENT, MAJORITY POLYFUNCTIONAL EPOXY RESIN,
HOT STRENGTH RETENTION, FLAME RESISTANT, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS.

Reinforcement material: G in accordance with MIL-S-13949.

Resin system. H in accordance with MIL-S-13949.

Special consideration (third letter designator): N or P in accordance with MIL-S-13949.

Type of metal foil: Any as specified in accordance with MIL-S-13949.

Weight of metal foil: H, M, 1, 2, 3, 4, 5 or 0 in accordance with MIL-S-13949.

Solderability: Base materials clad with metal foil types A, B, C, D, G, H, J, K, L, and M shall be tested in accordance with MIL-S-13949. Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist: C or X in accordance with MIL-S-13949.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method

Property values:

Flammability:

Duration of burning time, each flame application:

Base material thicknesses greater than or equal to .020 inch (0.51 mm).

Average: 25 seconds maximum

Single: 30 seconds maximum

Base material thicknesses less than .020 inch (0.51 mm): 15 seconds maximum

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:38Z
Check the source to verify that this is the current version before use.

Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm)).

Crosswise: 50,000 pounds per square inch minimum.

Lengthwise: 60,000 pounds per square inch minimum

Lengthwise at elevated temperature. 30,000 pounds per square inch minimum

Peel strength values for copper foil See table I

TABLE I. Peel strength values, minimum pounds per inch of width.

Test and copper foil weight	Base material thickness	
	Less than 0.020 inch	0.020 inch and over
After thermal stress:		
1/2-oz/ft^2 (designator H)	5.0	6.0
3/4-oz/ft^2 (designator M)	5.0	7.0
1-oz/ft^2 (designator 1)	6.0	8.0
2-oz/ft^2 (designator 2)	8.0	11.0
3, 4, or 5 -oz/ft^2 (designator 3, 4, or 5)	10.0	12.0
At elevated temperature of 150°C.		
1/2-oz/ft^2 (designator H)	4.0	5.0
3/4-oz/ft^2 (designator M)	4.5	5.0
1-oz/ft^2 (designator 1)	5.5	3.0
2-oz/ft^2 (designator 2)	5.0	4.0
3, 4, or 5 -oz/ft^2 (designator 3, 4, or 5)	8.0	5.0
After exposure to processing solutions		
1/2-oz/ft^2 (designator H)	4.0	5.0
3/4-oz/ft^2 (designator M)	5.5	5.0
1-oz/ft^2 (designator 1)	6.0	7.0
2-oz/ft^2 (designator 2)	6.0	9.0
3, 4, or 5 -oz/ft^2 (designator 3, 4, or 5)	8.0	12.0

Arc resistance 90 seconds minimum

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 40 kilovolts minimum.

Electric strength, average, (for base material thicknesses less than .020 inch (0.51 mm)): 750 volts/mil minimum

Loss tangent at 1 MHz. 0.035 maximum.

Permittivity at 1 MHz: 5.4 maximum.

Surface resistivity.

After moisture resistance conditioning 10^8 megohm minimum.

After elevated temperature (150°C) conditioning: 10^5 megohm minimum.

Volume resistivity

After moisture resistance conditioning: 10^7 megohm-cm minimum.

After elevated temperature (150°C) conditioning. 10^5 megohm-cm minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:38Z
Check the source to verify that this is the current version before use.

Water absorption.

Not applicable for base material thickness less than .020 inch (0.51mm).
Base material thickness of .020 inch (0.51 mm) to .031 inch (0 79 mm)· 0 80 percent maximum
Base material thickness of .032 inch (0.81 mm) to .062 inch (1.58 mm): 0.35 percent maximum
Base material thickness of .063 inch (1.60 mm) to .093 inch (2.36 mm): 0 25 percent maximum.
Base material thickness of .094 inch (2.38 mm) to .125 inch (3.18 mm): 0.20 percent maximum.
Base material thickness of .126 inch (3.20 mm) to .250 inch (6.35 mm): 0.13 percent maximum.

Supersession information: Not applicable.

Type designation· Consists of "GH" prefix followed by coded alphas and numeric characters as shown in the following example

GH N 0620 P3/P3 B 5 C

Base material
Special considerations (N or P, see MIL-S-13949)
Nominal base thickness (see MIL-S-13949)
Type of metal foil (side 1, see MIL-S-13949)
Weight of metal foil (side 1, see MIL-S-13949)
Type of metal foil (side 2, see MIL-S-13949)
Weight of metal foil (side 2, see MIL-S-13949)
Grade of pits and dents (A, B, C, or D; see MIL-S-13949)
Class of base material thickness tolerance (1, 2, 3, or 5, see MIL-S-13949)
Class of bow and twist (C or X; see MIL-S-13949)

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities·
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES
NSA - NS

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0024-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:38Z
Check the source to verify that this is the current version before use.